



FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestones 1

Due: May 19, 2026

Milestone	Description	Due Date
1	Capstone project selection/proposal	May 19
2	Literature review and data source identification	May 26
3	Data collection & Initial EDA	June 9
4	Data Preprocessing & Baseline Modeling	June 23
5	Advanced Modeling & Optimization	July 7
6	Validation & Business Impact Assessment	July 21
7	Capstone project oral presentation	July 27 & August 10
8	Capstone project report	August 10

In this course, students will apply the knowledge and skills acquired throughout the MBAI program to a major capstone project in business analytics and AI focused on a real-world business challenge. The project requires the integration of theoretical concepts with the creative application of AI technologies to address practical business problems.

The capstone project is structured as a guided experience in which students advance through a series of milestones and deliverables, culminating in a **final presentation** and **comprehensive report**. The report will detail their literature review, methodology, findings, proposed solutions, and the practical impact of their work, showcasing their ability to apply business analytics and AI effectively in real-world contexts.

The capstone project is expected to be completed in **groups of two**, although exceptions may be granted for students who can **significantly benefit** from working individually. **Should you wish to express a preference for your group members, use the self-sign-up link available on Canvas (Click on the People tab, then Groups) to join a group by Friday, May 15, 11:59 PM. Students without a group will be randomly grouped by this deadline, and switching groups will not be allowed after this date.**

Given the extensive scope and activities involved in a major project, the first milestone is essential for identifying a feasible project and obtaining high-quality, readily usable data that aligns with the available computing resources (e.g., laptops with limited RAM or free-tier cloud platforms such as Google Colab). A detailed breakdown is provided below:

Milestone 1: Capstone Project Selection/Proposal

Milestone 1 is designed to establish the foundation for the capstone project by defining a relevant business analytics or AI problem, identifying a feasible technical solution, and developing a structured project proposal supported by academic literature and data-driven justification. Students are expected to demonstrate graduate-level research, analytical thinking, technical planning, and professional writing.

The proposal should clearly explain:

- What business problem is being solved,
- Why the problem is important,
- How AI/analytics methods will be used,
- What data and tools will support the project,
- What measurable outcomes are expected, and
- How the project aligns with practical computing and time constraints.

Students must identify at least one current peer-reviewed research article published in 2025 or 2026 that will serve as the base paper for comparative analysis in the final capstone project. The selected article should directly relate to the proposed topic, methodology, or business problem.

1. Identify a Real-World Business Problem

Select a contemporary business challenge that can be addressed using:

- Artificial Intelligence (AI),
- Machine Learning (ML),
- Data Analytics,
- Business Intelligence (BI),
- Predictive Analytics,
- Natural Language Processing (NLP),
- Optimization techniques, or
- Generative AI applications.

Examples may include:

- Customer churn prediction,
- Fraud detection,
- Demand forecasting,
- Supply chain optimization,
- Employee attrition analysis,
- Financial risk assessment,
- Machine learning models to analyze historical financial market data
- AI-powered recommendation systems,
- Healthcare analytics,
- ESG and sustainability analytics,
- Predictive maintenance,
- Social media sentiment analysis.

The problem must:

- Have measurable business value,
- Be feasible within one academic term,
- Be supported by accessible datasets,
- Allow implementation using graduate-level analytical methods.

2. Refine the Problem Statement Using SMART Criteria

The project problem statement must satisfy SMART criteria:

SMART Element	Requirement
Specific	Clearly define the business issue
Measurable	Include quantifiable metrics
Achievable	Feasible with available tools/data and time
Relevant	Business or industry significance
Time-bound	Achievable within course timeline

Example:

“Develop a machine learning-based customer churn prediction model for the telecommunications industry capable of achieving at least 85% classification accuracy using Python and Scikit-learn on a dataset under 1GB within a 12-week development timeline.”

3. Specify Technical and Computing Constraints

Students must demonstrate awareness of project limitations and implementation feasibility.

Constraints may include:

- Dataset size limitations (recommended: under 1GB),
- Local hardware limitations,
- Cloud computing restrictions,
- Time constraints,
- Data availability,
- API access limitations,
- Model training limitations.

Example:

“The project will use publicly available datasets under 1GB to ensure compatibility with local computing environments without requiring GPU-based cloud infrastructure.”

4. Advisor Meeting

When uncertain about their project direction, students are encouraged to:

- **Schedule a meeting** with the instructor to discuss the feasibility of the proposed project.
- **Conduct a clarity check** to ensure the project aligns with course expectations and available resources.
- **Refine the project scope** after consultation to ensure:
 - Completion within the academic timeline,
 - Appropriate technical depth,
 - Clearly defined deliverables, and
 - Availability of the required datasets and tools.

Milestone 1 Deliverables

Students must submit one professionally formatted proposal document (**PDF** or **Word**) containing the following sections:

1. Title Page

The title page should include:

- Project Title
- Student Group Name
- Full Names of Team Members

2. Executive Summary / Abstract (100–200 words)

The abstract should briefly summarize:

- Business problem,
- Proposed AI/analytics solution,
- Methodology,
- Dataset and tools,
- Expected outcomes,
- Constraints and limitations.

The abstract should be concise, technical, and professionally written.

3. Problem Statement & Project Objectives

This section must include:

- **Formal Problem Statement**

Example:

“Develop a machine learning-based predictive analytics model to forecast customer churn in the telecommunications industry with a target accuracy of 85% using Python, Scikit-learn, and publicly available datasets.”

- **SMART Analysis**

Explain how the project satisfies SMART criteria.

- **Key Objectives**

Use measurable bullet points.

- **Base Research Article**

Students must identify:

- One recent peer-reviewed 2025 or 2026 article,
- Explain why it was selected,
- Describe how the project will compare or extend the findings.
- **Example:** Mitigating class imbalance in churn prediction with ensemble methods and SMOTE

4. Project Scope

Clearly define the scope of your project by specifying both the included and excluded components.

- **Included Components**

Your project scope should identify the following elements, where applicable:

- Algorithms and models

- Datasets
- Analytical techniques
- Visualization tools
- Evaluation metrics
- Programming languages and frameworks
- **Excluded Components**
Clearly state any features or functionalities that are outside the scope of the project.
Examples may include:
 - Real-time deployment
 - Production-level APIs
 - Enterprise system integration
 - Mobile application development
 - Large-scale distributed computing

5. Data Sourcing Plan

Students must clearly identify the following:

- The primary dataset source to be used for the project
- One or more backup datasets in case the primary dataset becomes unavailable, incomplete, or unsuitable for the project requirements

6. Risk Assessment & Mitigation Plan

Students should identify potential risks that may impact the success of the project and provide appropriate mitigation strategies for each identified risk.

Examples of potential risks include:

- Insufficient or poor-quality data
- Model overfitting
- Computational or hardware limitations
- Scope creep
- Time management and scheduling challenges

7. References (IEEE Style)

Students must use proper IEEE citation formatting consistently throughout the proposal.

Example References

[1] M. G. Abdelhady and K. A. Mohamed, "Leveraging artificial intelligence for predictive customer churn modeling in telecommunications: A framework for enhanced customer relationship management," *Scientific Reports*, vol. 15, 2025.

[2] S. Kimitei *et al.*, "Predictability and explainability of survival analysis in churn prediction," *Journal of Marketing Analytics*, 2025.

Submission Guideline:

Please submit one proposal per group through Canvas no later than **May 19, 2026**, at 11:59 PM. The proposal submitted for Milestone 1 will serve as the foundation for the final capstone report. It is expected that the proposal may evolve throughout the course based on instructor feedback and ongoing project development.